```
clc;
clear;
close all;

% FIR Windowing Techniques Comparison

N = 50;             % Filter order
Fs = 8000;           % Sampling frequency
fc = 1500;           % Cutoff frequency in Hz
wc = fc/(Fs/2);      % Normalized cutoff frequency

% Window functions
w_rect  = rectwin(N+1);
w_hamm  = hamming(N+1);
w_hann  = hann(N+1);
w_black = blackman(N+1);

% FIR low-pass filter design
b_rect  = fir1(N, wc, w_rect);
b_hamm  = fir1(N, wc, w_hamm);
b_hann  = fir1(N, wc, w_hann);
b_black = fir1(N, wc, w_black);

% Frequency responses
[H_rect, f] = freqz(b_rect, 1, 1024, Fs);
[H_hamm, ~] = freqz(b_hamm, 1, 1024, Fs);
[H_hann, ~] = freqz(b_hann, 1, 1024, Fs);
[H_black, ~] = freqz(b_black, 1, 1024, Fs);

% Figure 1: Magnitude and phase responses
figure(1);

subplot(2,1,1);

plot(f, 20*log10(max(abs(H_rect),1e-10)), ...
    'k', 'LineWidth', 1);
hold on;

plot(f, 20*log10(max(abs(H_hamm),1e-10)), ...
    'b', 'LineWidth', 1.2);

plot(f, 20*log10(max(abs(H_hann),1e-10)), ...
    'g', 'LineWidth', 1.2);

plot(f, 20*log10(max(abs(H_black),1e-10)), ...
    'r', 'LineWidth', 1.2);

grid on;
xlim([0 Fs/2]);
ylim([-100 10]);

title('FIR Low-Pass Filter Magnitude Response (N = 50)');
```

```
xlabel('Frequency (Hz)');
ylabel('Magnitude (dB)');

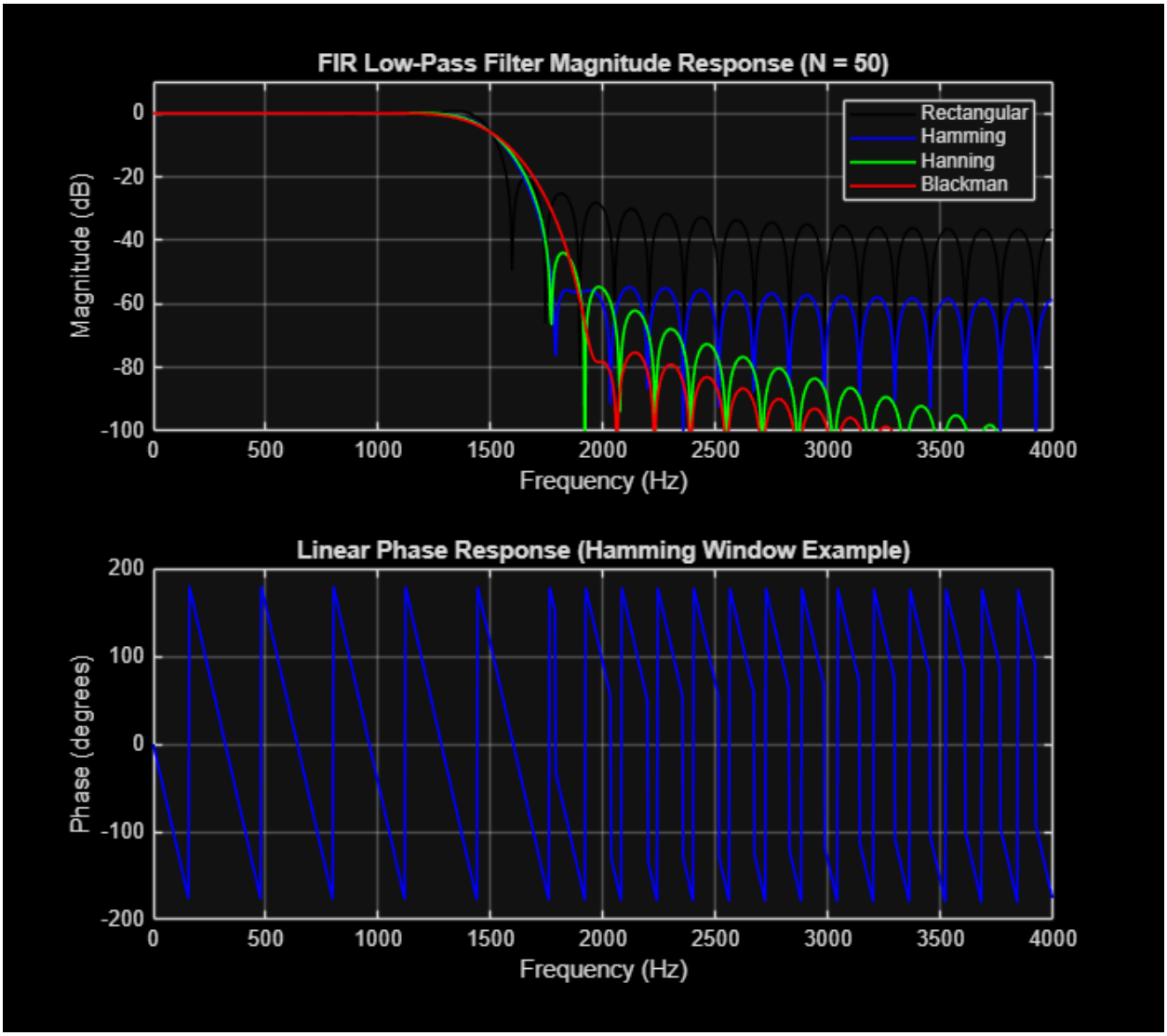
legend('Rectangular','Hamming','Hanning','Blackman', ...
    'Location','northeast');

% Hamming window phase response
subplot(2,1,2);

plot(f, angle(H_hamm)*180/pi, ...
    'b', 'LineWidth', 1.2);

grid on;
xlim([0 Fs/2]);
ylim([-200 200]);

title('Linear Phase Response (Hamming Window Example)');
xlabel('Frequency (Hz)');
ylabel('Phase (degrees)');
```



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